Lecture: 01
11th September 2026

المعهد الهندي للتكنولوجيا دلهي
أبوظبي

INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

ASBL100 Semester-I 2026

Introductory Biology For Engineers

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Why you need to study Biology?

Biology is very ……..subject.

- **Our goal is to make you (Engineering students) learn and understand HOW BIOLOGY WORKS**

- **In contrary Biology students wish to understand WHAT BIOLOGY IS.**

- Goal is to make aware about basic biology concepts.
- You can apply your knowledge to novel situations.
- You can find motivation to contribute in various researches in biology.
- e.g. genetic engineering, biomedical devices, biological machines

Information transfer in organisms is like information transfer in a computer



Hard disk

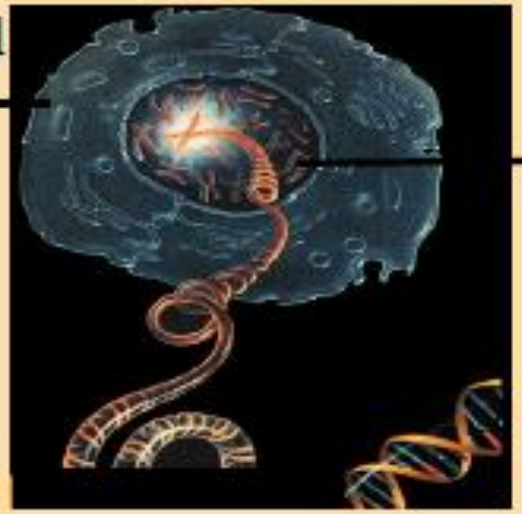


Files





Cell



Genome

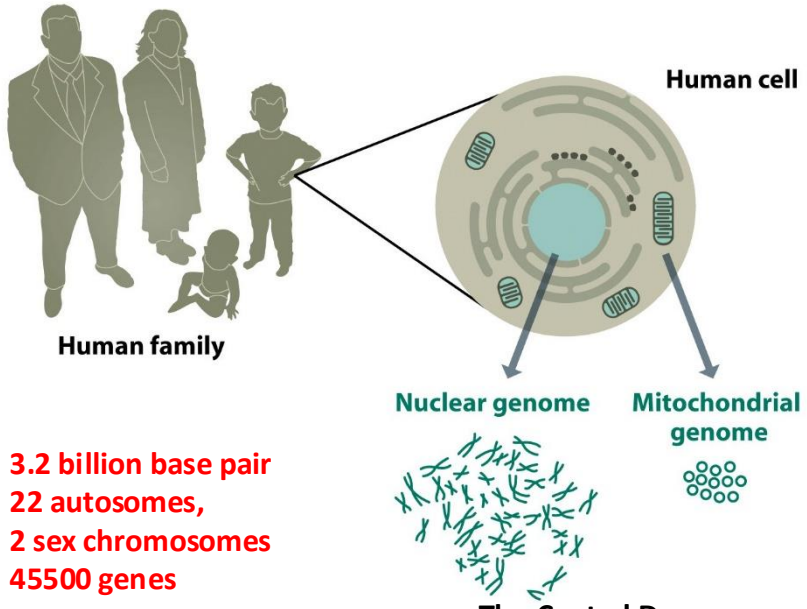


Chromosome

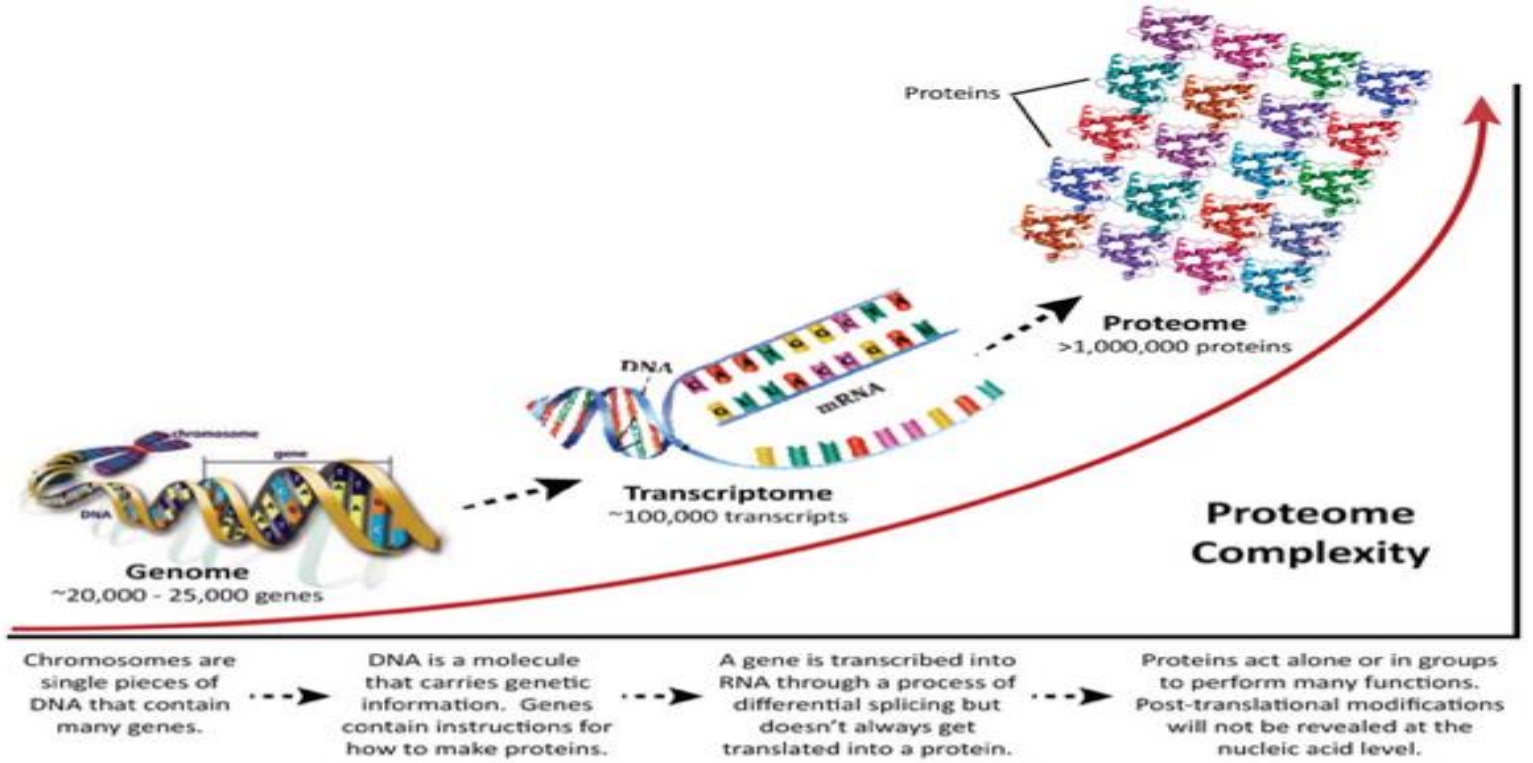


Gene

Complexity



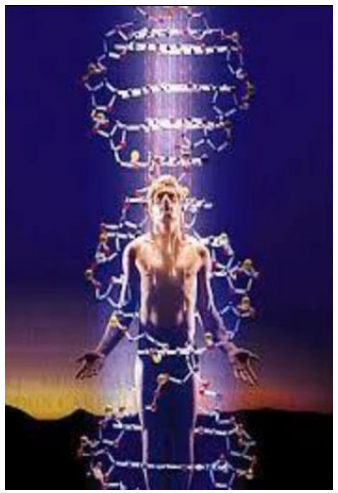
3.2 billion base pair
22 autosomes,
2 sex chromosomes
45500 genes

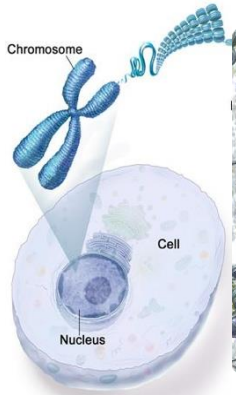


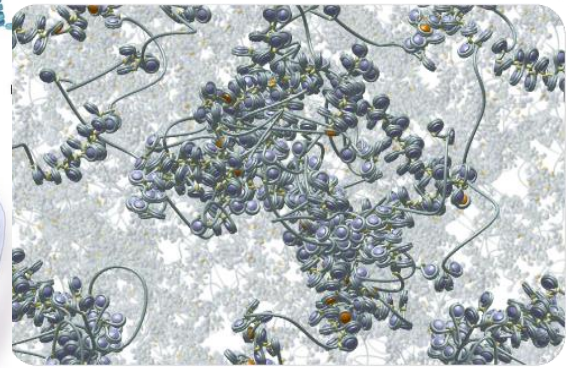
The Central Dogma

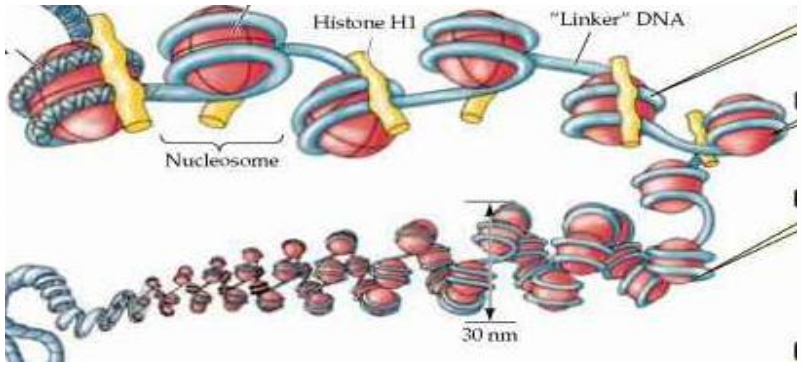
Transcription → Translation

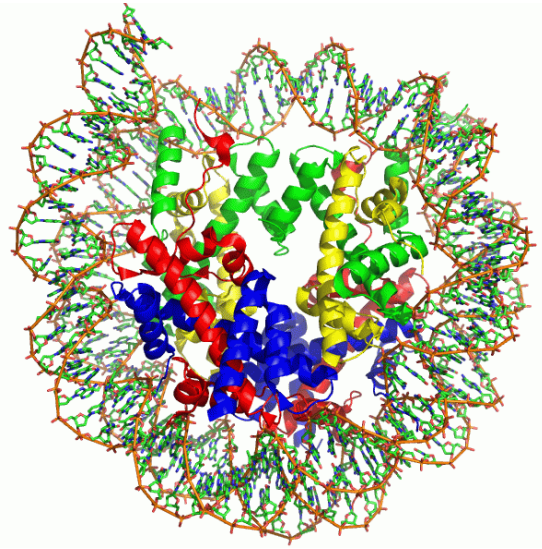
DNA → RNA → Protein → Trait





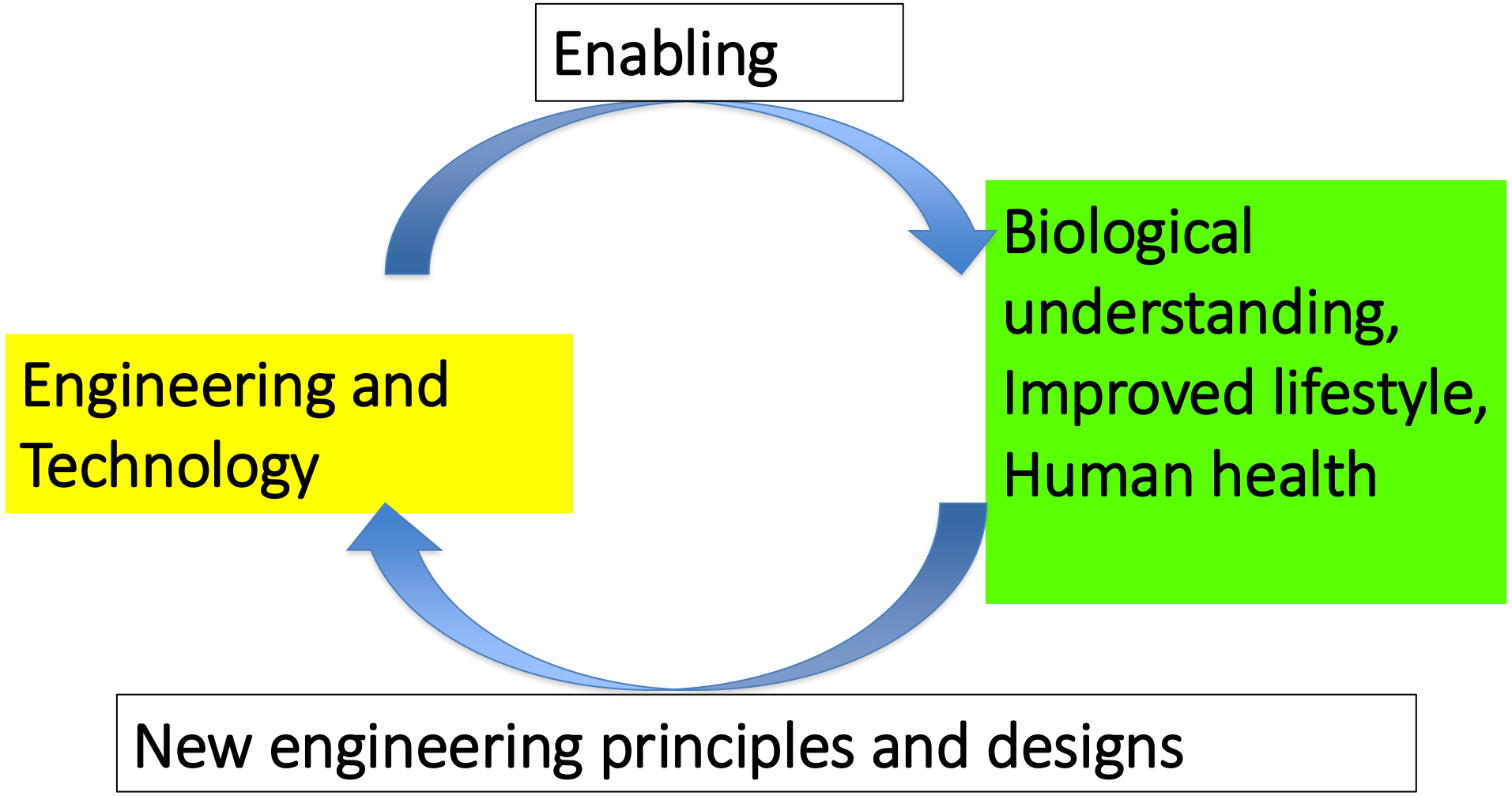






Each of the ~10^{13} cells in the human body receives lakhs of DNA lesions per day. These lesions can block genome replication and transcription, and if they are not repaired or are repaired incorrectly, they lead to diseases.

Relation between Biology and Engineering Technology



Need Bright Engineers with great ideas, excitement, and motivation students from chemistry, computer science, math, physics and engineering have equal chance to contribute in biology

Engineering can lead to high impact development for advancement in quality of life

Biomedicine

Complex molecular devices for tissue repair/regeneration

Smart drugs: directly sensing of molecular disease and making a diagnostic decision, then translated into drug activation.

Biological delivery systems

Vectors for therapy

Personalized medicine

Cells with new properties that improve human health

USE OF BIOLOGY IN ENGINEERING

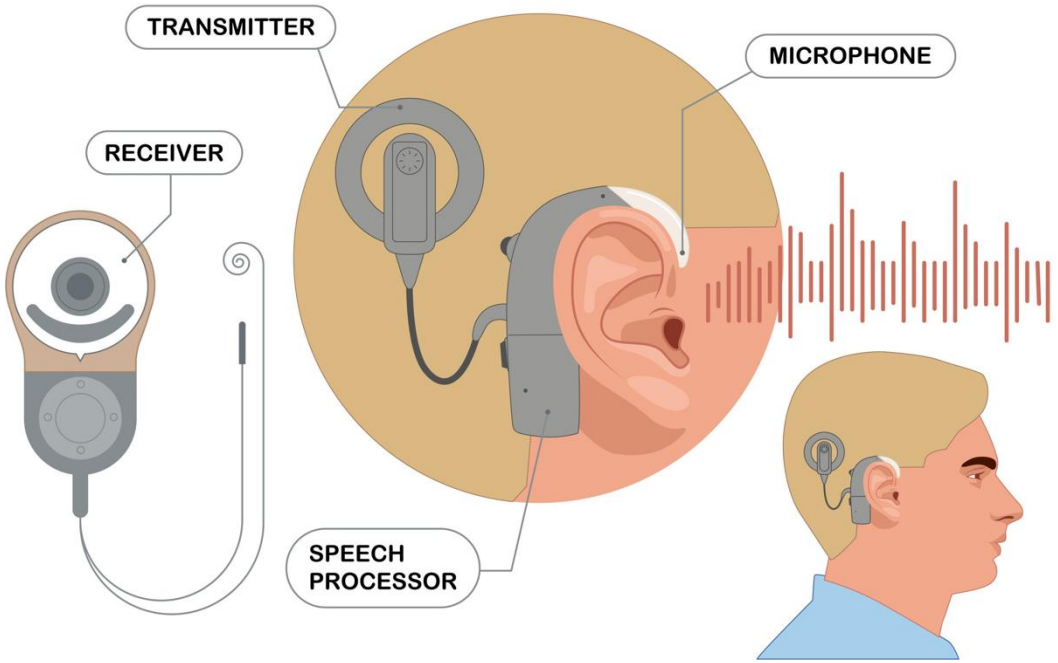
- Mechanical Engineering: Biological system prosthetics.
- Computer science and engineering: Computational biology robotics.
- Electrical engineering: Robotics devices and photonics signals & image processing.
- Chemical engineering: Biotechnological nanoparticles interfaces polymers biofuels.

Bio-Inspired Technologies

Engineering knowledge has driven breakthroughs in biomedical applications, creating job opportunities in a rapidly growing industry.

- **Designing Advanced Medical Devices**

The global cochlear implant market is valued at approximately $4.15 billion, projected to reach $6.24 billion by 2031



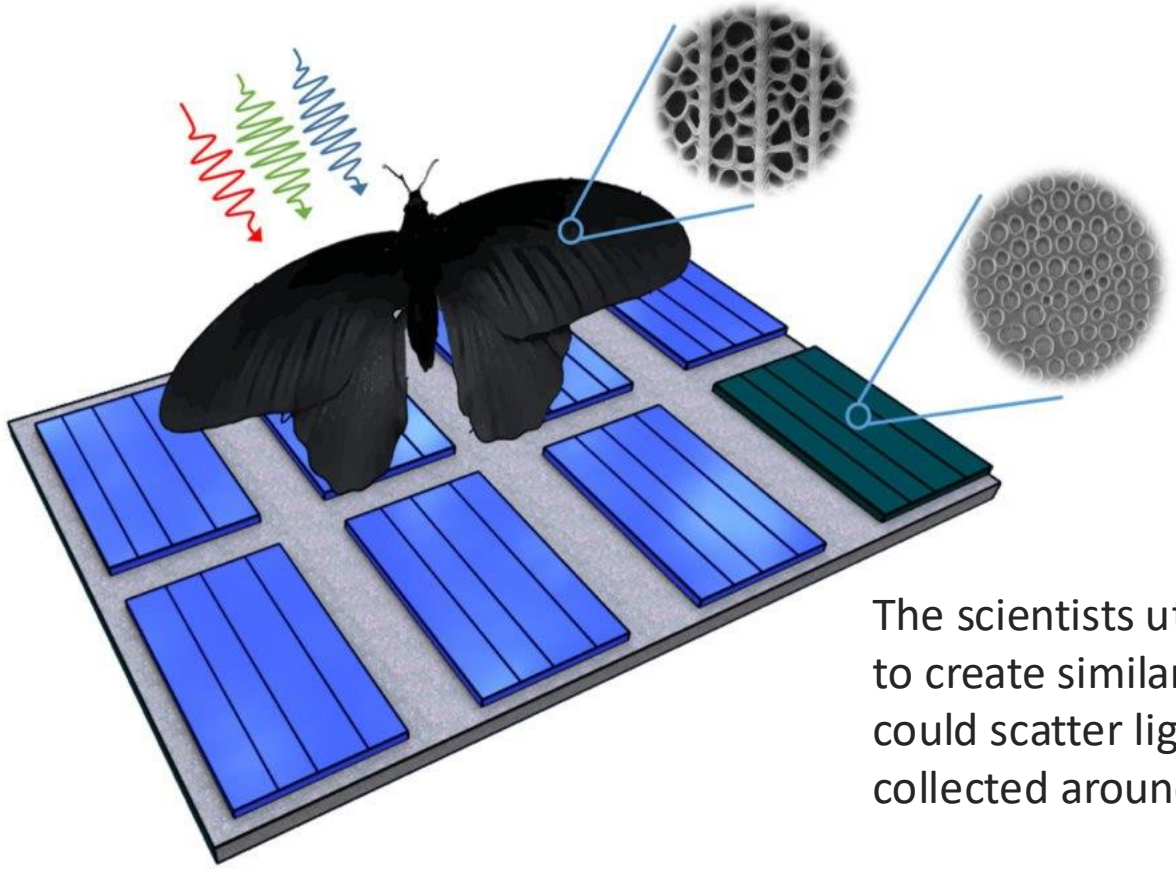
Cochlear Implant

The global prosthetic limb industry is valued at over $7 billion,



Prosthetic Limb

California Institute of Technology and Karlsruhe Institute of Technology researchers went to nature for inspiration for improved solar power technology. They scrutinized butterfly wings under an electron microscope and discovered the wings' scales were pockmarked with holes, which are under one millionth of a meter wide, according to The Verge. These holes not only allowed the wings to be lighter, but scattered light so the butterfly can absorb more heat.





Standard solar panels only peak at direct noon angles. Butterfly-inspired solar cells boost absorption of early morning, late evening, and cloudy diffuse light by up to double

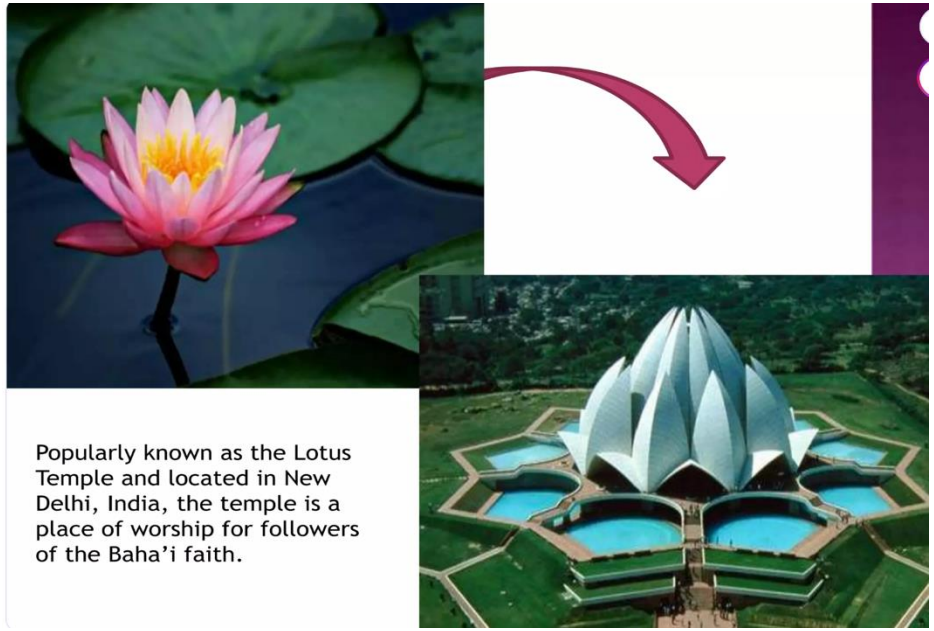
The scientists utilized hydrogenated amorphous silicon sheets, to create similar structures. A top layer had small holes that could scatter light, allowing it to hit the silicon base. This design collected around twice as much light as others.

The holes are random in shape, size, and distribution, according to Radwanul Siddique, lead author of a study recently published online in the journal *Science Advances*. Computer models helped the scientists realize the shape wasn't important for absorbing light, but position and order did matter.

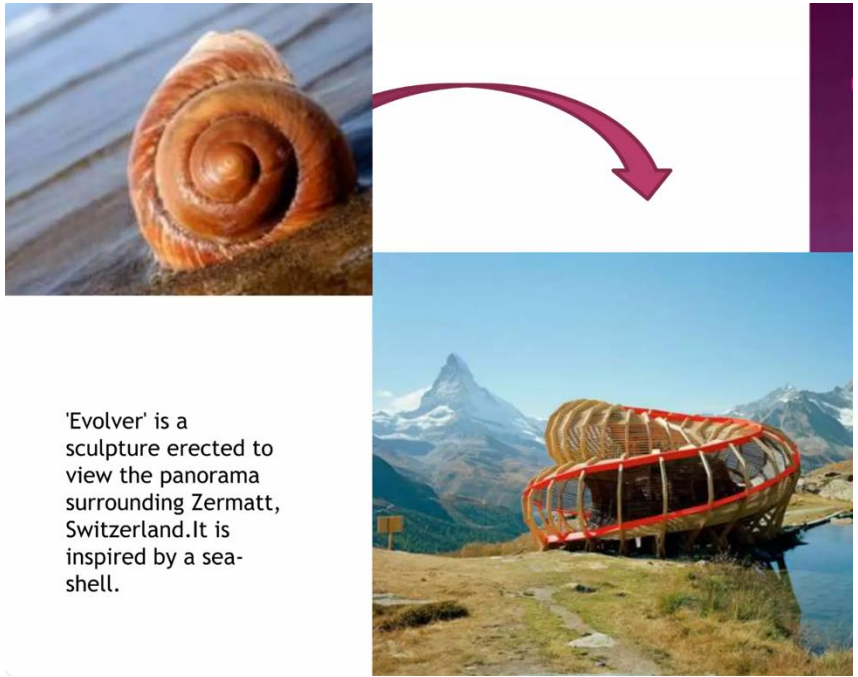
Some examples of Bio-inspirations



The Palm Jumeirah islands are artificial islands built on the coast of Dubai.Inspired by palm trees, the islands are in the shape of a palm tree.



Popularly known as the Lotus Temple and located in New Delhi, India, the temple is a place of worship for followers of the Baha'i faith.



'Evolver' is a sculpture erected to view the panorama surrounding Zermatt, Switzerland.It is inspired by a sea-shell.



The Quadracci Pavilion, designed by is inspired by a bird's wings, and contains a movable sun-breaker in the shape of wings that open up during the day and fold over the arched structure at night.

The Quadracci Pavilion is a famous postmodern landmark addition to the Milwaukee Art Museum in Wisconsin, USA.

Key Features

- **Architect:** Designed by the renowned Spanish architect and engineer Santiago Calatrava.

- **Brise Soleil:** Features a movable sunscreen with a 217-foot (66-meter) wingspan that opens and closes like a bird.

The toes of the gecko have inspired adhesive that is strong enough to allow a human to climb up a glass wall.



BIOMIMETICS:BIO-INSPIRED TECHNOLOGIES

- Aircraft wing design: inspired by birds and bats
- Bio robots: based on the physiology and methods of locomotion of animals
- Camera: inspired by human eye
- Pheromone Sensors: Ant-Inspired

Biomimicry: Emulating nature's functions, mechanisms, and strategies to create sustainable designs.

Bioinspired: Drawing general inspiration or ideas from biological systems without strictly copying them





In 1941, a Swiss engineer George de Mestral was hunting with his dog when he realized small burrs from the burdock plant were stuck to his dog's hair. On closer inspection, he realized that these tiny hooks and loops can be made into clothing or garment fasteners which led to the discovery of velcro





- **Peacock-Inspired Biosensors**
- Engineers at the University of Surrey and the University of Sussex have found a way to mimic these patterns using graphene to produce opal-like photonic crystals that can change color in response to not just light, but temperature, chemical changes, strain and more.
- The crystals can be used for a number of medical applications, including biosensing technologies to detect respiratory viruses; bio-monitoring to improve or enhance physical performance; or even healthcare safety to determine how sanitary a surface is.

MIT "Russian Doll" teams were awarded a total of **$23 Million to fight brain tumors**

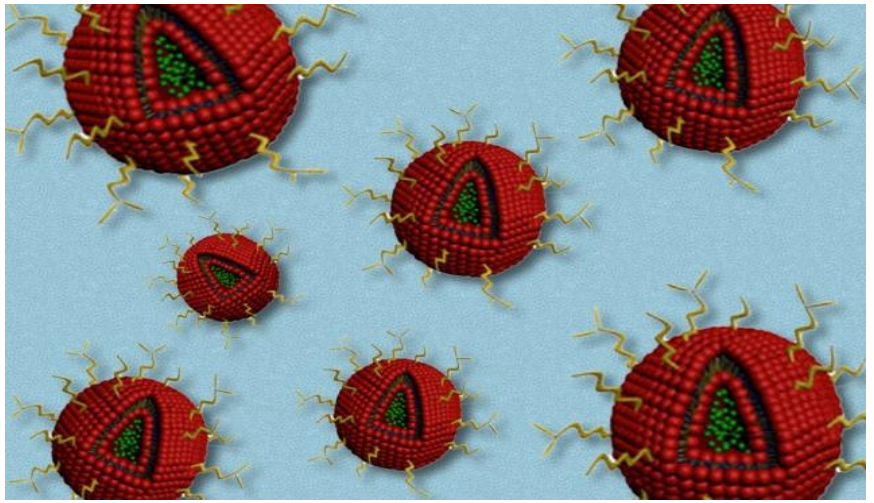
Researchers from MIT's Koch Institute will work with teams in the UK and Europe to use nanoparticles to carry multiple drug therapies to treat glioblastoma. The team will use tiny "Russian doll-like" particles — a technology developed at MIT — to deliver drug combinations to brain tumors



Nested Layers: Inspired by traditional wooden nesting dolls, these synthetic nanoparticles feature concentric layers designed to hold and release distinct therapeutic payloads sequentially

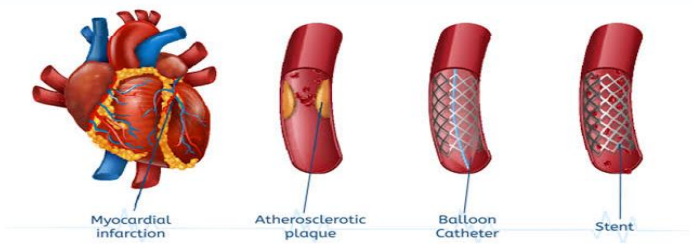
Temozolomide, which is usually the first chemotherapy drug given to glioblastoma patients.

Bromodomain inhibitors JQ-1 are believed to interfere with cells' ability to repair DNA damage



they coated the liposomes with a protein called transferrin, the particles could pass through the blood-brain barrier with little difficulty. Furthermore, transferrin also binds to proteins found on the surface of tumor cells, allowing the particles to accumulate directly at the tumor site while avoiding healthy brain cells

few familiar examples of what can result when engineering knowledge is applied to solving problems in medicine and biology:



- Cardiac stents
- Defibrillators
- EKGs
- Holter monitors
- Mobile Cardiac Outpatient Telemetry (MCOT)
- EEGs
- Medical imaging (x-ray, CT, MRI, fMRI, PET)
- The cochlear implant
- Anesthesia monitoring equipment
- Prescription monitoring for pharmacies
- Artificial hearts and valves
- Pacemakers
- Medical ventilator systems
- Rehabilitation systems
- Prosthetics
- LASIK surgery
- daVinci surgical robots
- Transcatheter valve replacement and repair devices

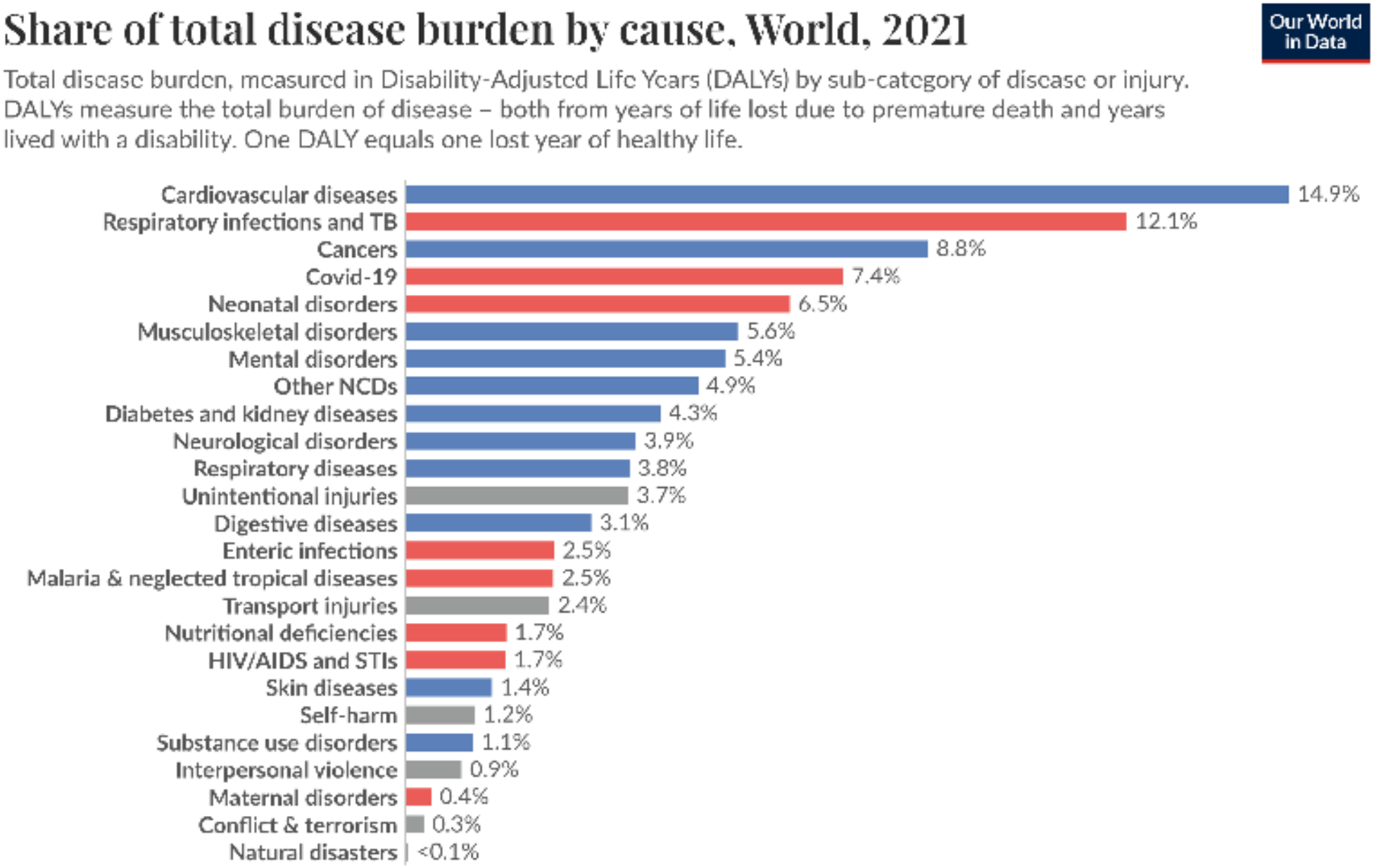
Advancements in telemedicine, stem cell research, nanotechnology, tissue engineering, wearable technologies for home health monitoring, and neural research can all be attributed to the work of biomedical engineers.

The result of Engineers are giving rise to such advancements as healthcare tools, intelligent drugs (nano particles) and replacement tissues.

With advanced computer models that provide the means for interpreting diagnostic values in new ways, engineering is moving us in the direction of more personalized therapy.

Beyond simply providing technical devices, engineers can help physicians to facilitate decision-making, thereby advancing the science of medicine and freeing physicians to focus on the art of patient care.

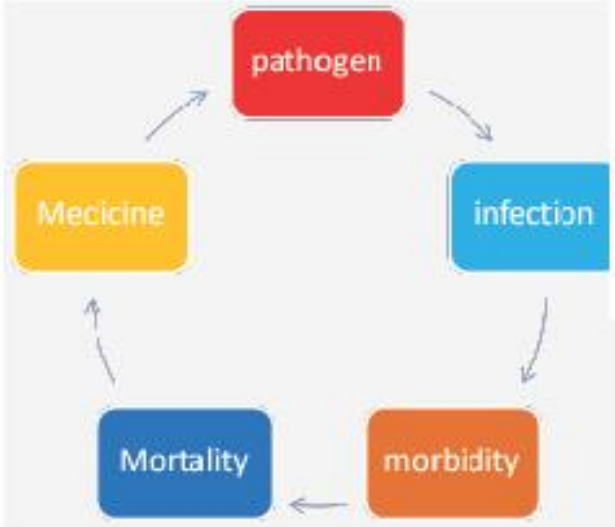
Potential opportunity for Engineering Technology in health

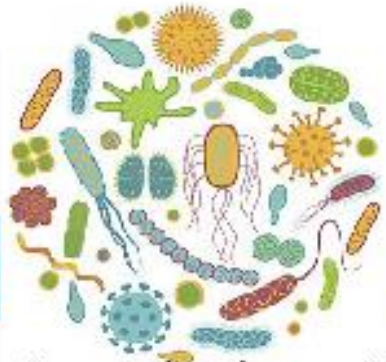


Problems needs to be addressed

Challenges







HIV/AIDS
polio common cold herpes
mumps diarrhoeal disease
Ebola rubella influenza
yellow fever meningitis rabies
measles hepatitis

Sources of infections are of size µm to nm

Cancer, Diabetes, Arthritis, Neuro-disorders, kidney, heart, Hypertension, stroke, eye, liver, viral diseases Chikungunya, Dengue, COVID-19 and many more



Unhealthy

Healthcare



Healthy



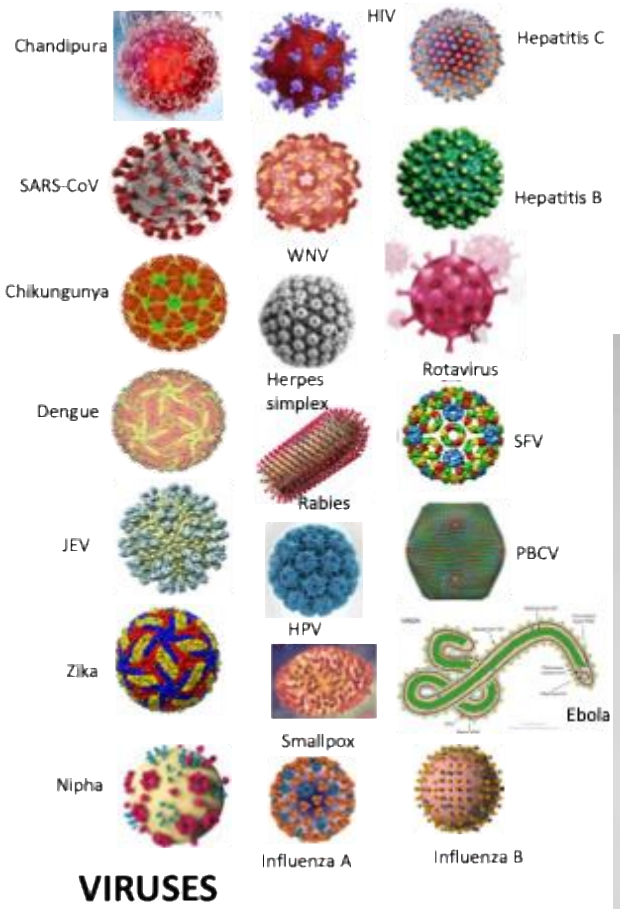






Modern medicine, Ayurveda, Yoga & Naturopathy, Unani, Siddha and Homeopathy

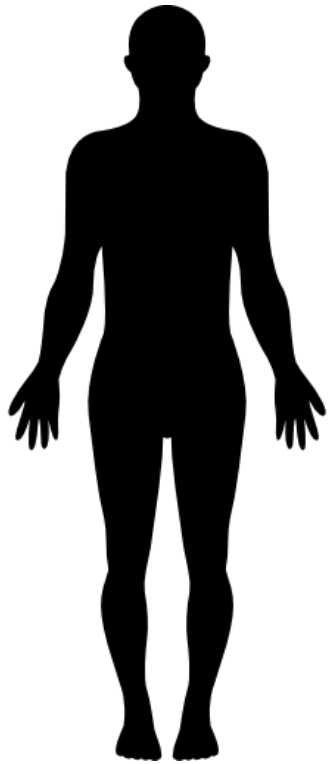
Who is smart?



Viruses are
helpless, homeless,
~100 nm in size,
~30000 bp genome
~picogram in weight

Small Viruses/pathogens
Big impact



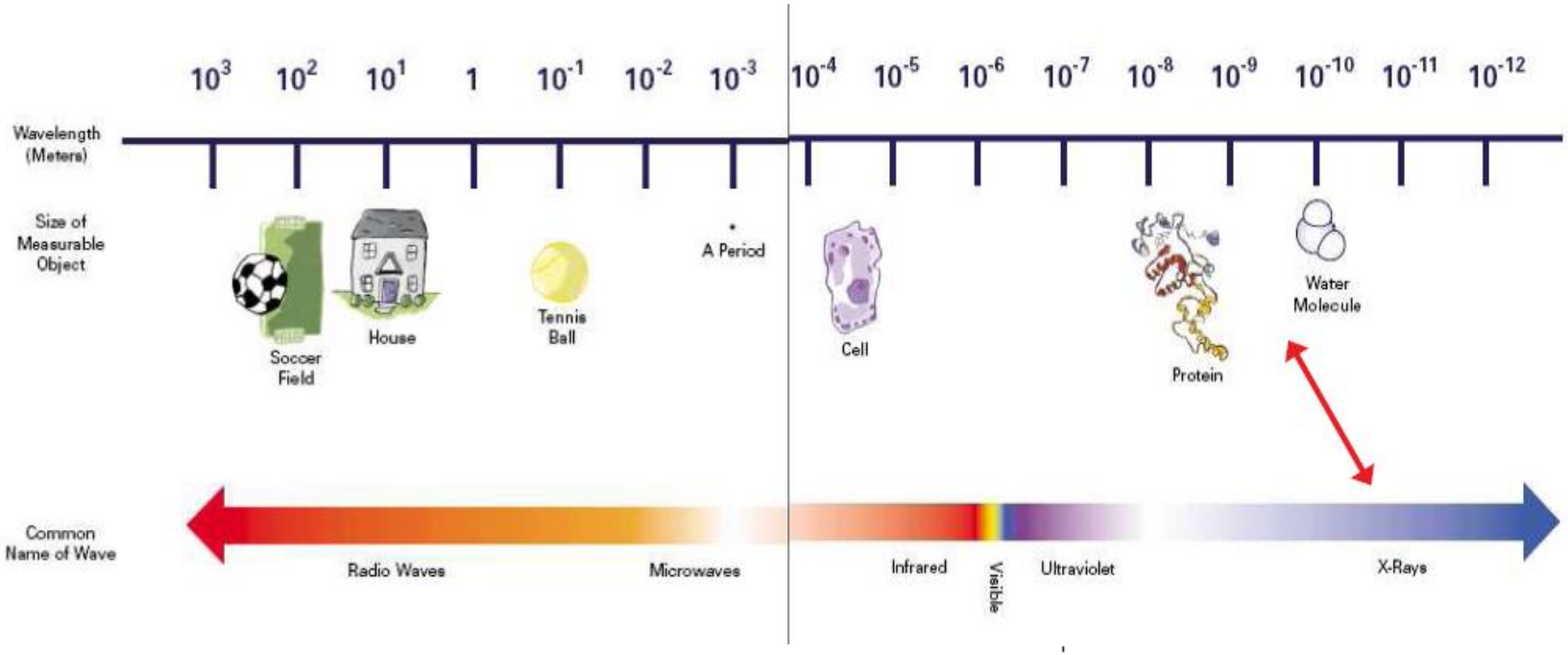


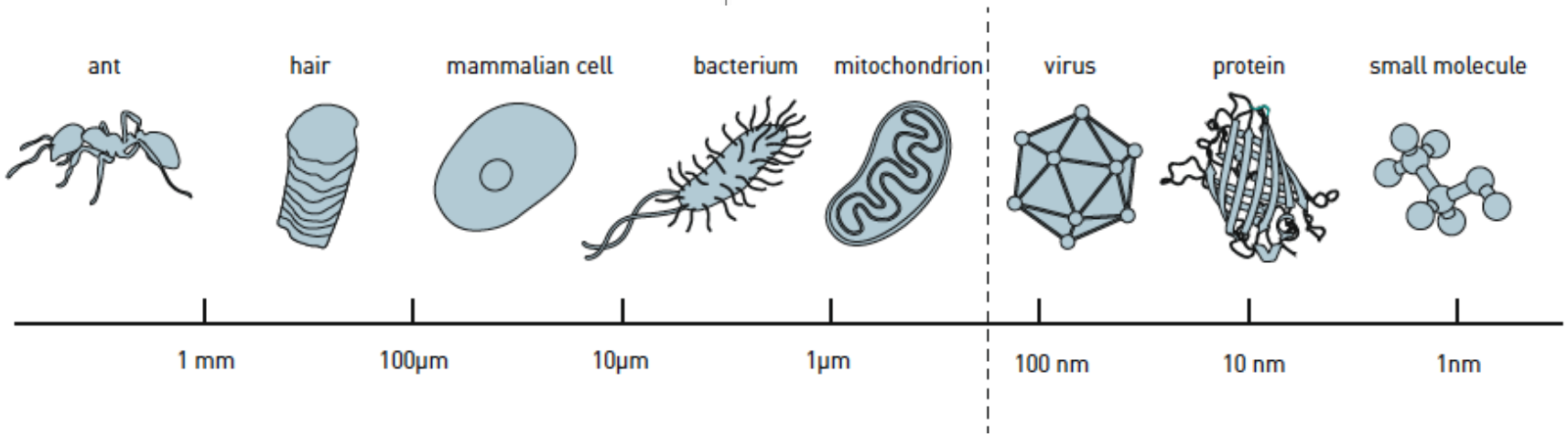
Human being,
AI intelligent,
3 BHK,
2 meter in size,
6.4 billion bp genome
~60 kg weight

Viruses are the Job provider to lakhs of researchers, doctors, hospitals.

Job seeker

Main reason in failure of our not understanding the biological problems is because the cells, proteins, DNA are very very tiny,





Advance techniques to see the structure of machines





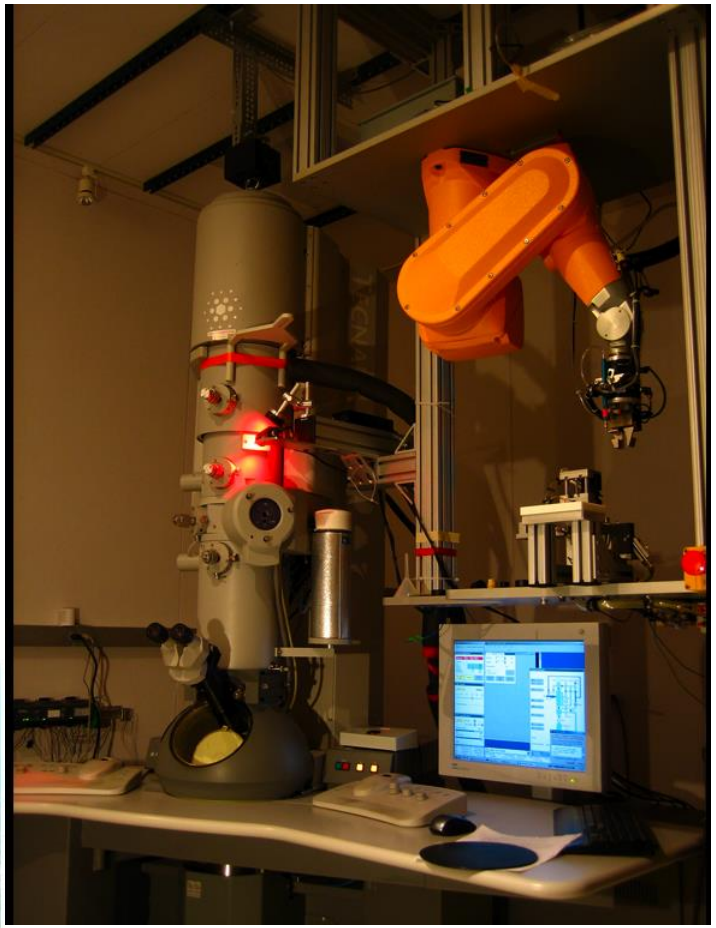
high-resolution equipment which can help in understanding Viruses and proteins

Cryo-electron microscope

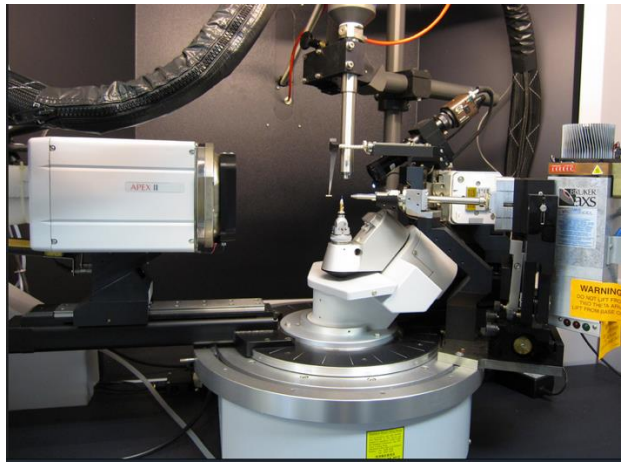
Noble Prize 2017 in chemistry
For discovery of Cryo-Electron Microscopy

Jacques Dubochet University of Lausanne in Switzerland, Joachim Frank at Columbia University, and Richard Henderson at the MRC United Kingdom

The wavelength of electrons is much shorter than that of light, it can reveal much finer detail



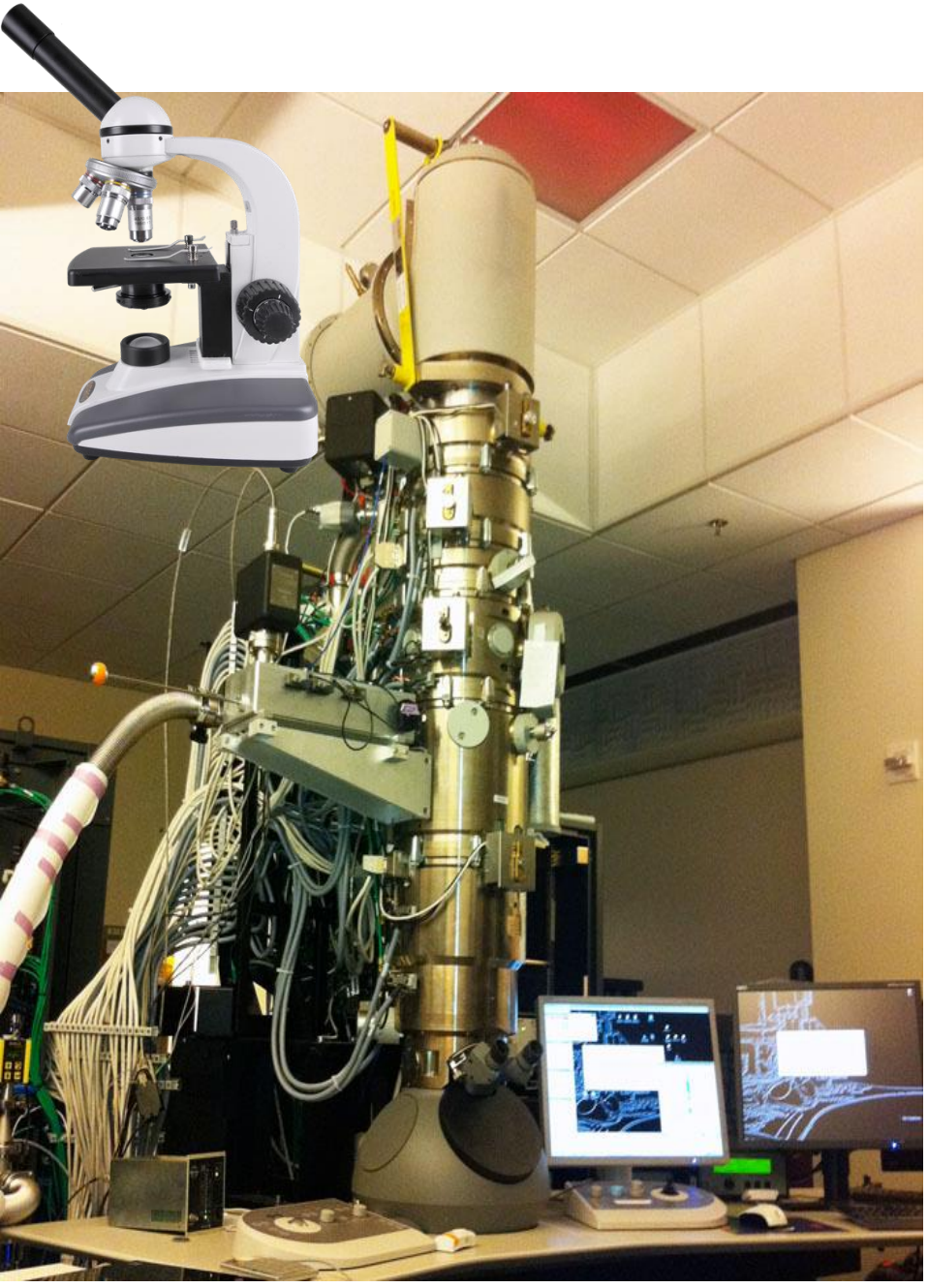


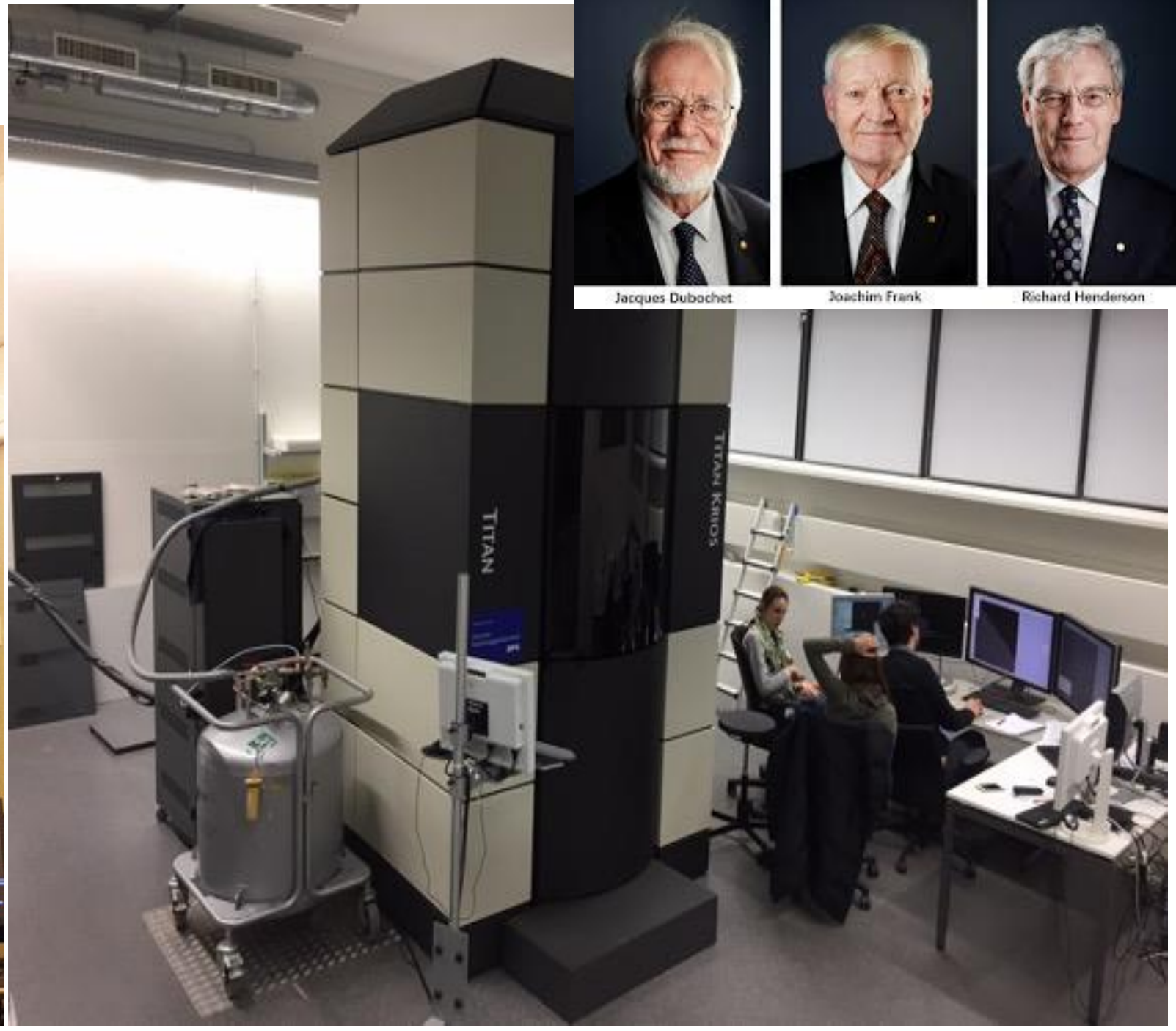




X-Ray diffractometer

Cryoelectron microscope





Jacques Dubochet

Joachim Frank

Richard Henderson

Any Questions/ Doubts?

Thanks for your Patience

Have a good day